

Explaining Frequency Domain CNN Predictions with Score-CAM: A Comprehensive XAI Approach for Image Classification

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Table of Contents

1. Introduction
2. Literature Review
3. Research Gaps Identified
4. Problem Statement
5. Objectives
6. Proposed Methodology
7. Implementation Details
8. Results and Analysis
9. Conclusion
10. Future Work
11. References

Introduction

- ▶ Traditional CNNs work well with spatial data but have trouble handling high-resolution and large images. Using frequency domain processing with FFT (Fast Fourier Transform) makes CNN training and use faster and more efficient in terms of memory.
- ▶ Explainable AI techniques, like Grad-CAM/Score-CAM, are commonly used to understand how models make predictions. They create visual explanations that show which parts of an image are most important for the model's decision.
- ▶ A major research question is whether XAI methods, such as Grad-CAM/Score-CAM, can be successfully used with CNNs that operate in the frequency domain?

Literature Review - FFT

Author	Description	Methodology	Results	Limitations
Vasilache et al.	FFT-based CNN convolution implementations using cuFFT and custom fbfft library for GPU performance optimization.	Implemented FFT convolution using cuFFT with autotuning strategy. Developed custom fbfft for batched 1-D/2-D FFTs.	cuFFT: 1.4x-14.5x speedup over cuDNN, up to 23.54x for 13×13 kernels. fbfft: 1.5x over cuFFT, 78% efficiency at 97.5% occupancy.	Poor performance for small 3×3 kernels due to overhead. fbfft limited to square, power-of-2 sizes (8-64 range).
Mathieu, Henaff, LeCun	Fast CNN training with FFTs using Fourier domain convolutions and efficient feature map reuse.	Transform inputs/kernels to frequency domain, perform element-wise multiplications, apply inverse FFT. Custom CUDA implementation with FFT reuse.	$2-3 \times$ speedup over Torch7/CudaConv. Training: 867ms vs 1756ms. Kernel-size independent with negligible accuracy loss (10^{-5} - 10^{-4}).	Extra GPU memory for frequency representations. Padding for non-power-of-2 inputs. High overhead for small networks.

Literature Review - FDCNN

Author	Description	Methodology	Results	Limitations
Xu, Qin, Sun, Wang, Chen, Ren	DCT-based CNN learning with frequency channel selection for image classification.	RGB→YCbCr→DCT, group into 192 frequency channels, gate module with Gumbel Softmax for dynamic selection.	ImageNet: +1.60% (ResNet-50), +0.63% (MobileNetV2). COCO: +0.8% AP. 87.5% channel pruning with minimal loss.	Fewer channels sometimes better (high-frequency noise). Limited frequency-spatial mapping analysis.
Goh, Chen, Lima	Frequency Domain Convolutional Neural Network (FDCNN) for large image classification using FFT-based convolution operations.	Proposed FDC/FDP layers with RFFT, spatial-to-frequency kernel transform, CIC, artifact removal. Built FDCNN and modified VGG16.	FDCNN: 54.21% faster training, 70.74% memory efficient. Full FDC: 46% faster at 256/512 channels.	Lack of suitable frequency domain activation function limits architecture design. FDCNN shows accuracy decrease when overfitting begins.

Literature Review - XAI

Author	Description	Methodology	Results	Limitations
Ribeiro, Singh, Guestrin	LIME explains black-box predictions by learning local interpretable models. Perturbs input data to understand feature importance.	LIME perturbs inputs locally, fits weighted sparse models to highlight features. SP-LIME uses submodular optimization for dataset explanations.	LIME provides faithful explanations for text/image classifiers. Shows improved trust, model selection, and spurious correlation detection.	Quality depends on model locality. Inadequate for nonlinear regions. High cost for complex models/images.
Scott M. Lundberg, Su-In Lee	SHAP: Shapley values for feature attribution in predictions. Unifies explanation methods under additive framework.	Additive feature attribution as linear sums. Shapley values satisfy accuracy, missingness, consistency axioms. Efficient approximations: Kernel SHAP, Deep SHAP.	SHAP outperforms LIME/DeepLIFT in accuracy and intuition. Kernel SHAP: higher efficiency, faster convergence. Validated on text/image models (MNIST).	Exponential in features; needs approximations. Assumes independence/linearity. High cost for large/correlated inputs.
Iftikhar, Anjum, Siddiqui, Rehman, Ramzan	Interpretable CNN for brain tumor detection using MRIs. Integrates XAI techniques (Grad-CAM, SHAP, LIME) to highlight abnormal tissues and improve transparency.	Two MRI datasets (Msoud: 7023, NeuroMRI: 3264) with preprocessing. CNN iteratively refined using Grad-CAM to remove layers. Explanations via Grad-CAM, SHAP, LIME.	XAI-CNN: 99.21% accuracy/F1 on main dataset, 94.72% on test set. Visualizations show focus on clinically relevant tumor features, outperforming non-XAI approaches.	Limited to MRI; may not generalize to CT/PET. Only tested Grad-CAM, SHAP, LIME. Reduced performance on unseen data due to domain variability.

Literature Review - Grad-CAM

Author	Description	Methodology	Results	Limitations
Selvaraju, Cogswell, Das, Vedantam, Parikh, Batra	Grad-CAM: Visual explanations for CNNs using gradients to create localization maps highlighting important regions for predictions.	Uses gradients from class scores to final conv layer for importance weights. Weighted activation maps with ReLU create localization maps. Combines with Guided Back-propagation for high-res visualizations.	ILSVRC15: 56.51% vs 61.12% error. Human studies: 61.23% vs 44.44%. 0.254 correlation with occlusion. Applied to classification/captioning/VQA. Identifies biases/failures.	Coarse resolution (14×14 for VGG/Alexnet). Needs other methods for high-res. ReLU misses negatives. Limited to decisions.
Chattopadhyay, Sarkar, Howlader, Balasubramanian	Grad-CAM++: Improved visual explanations for CNNs with better object localization and multiple instance handling via pixel-wise gradient weighting.	Weighted positive derivatives of final conv layer w.r.t. class scores. Pixel-wise weights via 2nd/3rd derivatives. Exponential/softmax for smooth differentiation. Combines with Guided Back-prop for high-res.	Outperformed Grad-CAM: 36.84% vs 46.56% drop, 70.72% win rate. Human preference: 109.69 vs 56.08. IoU: 0.38 vs 0.28. Applied to captioning/3D/distillation (+35.5% mAP). Works on VGG-16/AlexNet/ResNet-50.	Higher-order derivatives increase complexity. Variable across architectures/datasets. Coarse maps need upsampling. Limited to positive gradients. Higher overhead than Grad-CAM.

Literature Review - Score-CAM

Author	Description	Methodology	Results	Limitations
Wang, Wang, Du, Yang, Zhang, Ding, Mardziel, Hu	Score-CAM: Gradient-free visual explanations using forward-passing scores to create class activation maps without relying on gradients.	Extract activation maps from the last convolution layer, upsample and normalize them to [0,1], apply element-wise masking on the input, perform forward passes to obtain class scores, normalize the scores as weights, and combine the weighted maps with ReLU to generate the final localization map.	Recognition: 31.5% average drop vs 45.5% (Grad-CAM++). 30.6% average increase vs 18.9% (Grad-CAM++). Localization: 63.7% energy-based pointing vs 49.3% (Grad-CAM++). Insertion AUC: 0.386 vs 0.346. Deletion AUC: 0.077 vs 0.082. Passes sanity check.	It is computationally expensive due to multiple forward passes, slower than gradient-based methods, requires high GPU memory, sensitive to baseline input choice, and works best only on the last convolution layer.

Base Paper Datasets

- ▶ *Learning in the Frequency Domain: ImageNet 2012 Large-Scale Visual Recognition Challenge (ILSVRC-2012) dataset, COCO dataset*
- ▶ *Frequency Domain Convolutional Neural Network: Accelerated CNN for Large Diabetic Retinopathy Image Classification: APTOS 2019 blindness detection competition dataset*
- ▶ *Explainable CNN for brain tumor detection and classification through XAI based key features identification: Msoud dataset, NeuroMRI dataset*

Base Paper Datasets

- ▶ *Visual Explanations from Deep Networks via Gradient-based Localization*: ImageNet localization challenge (ILSVRC-15) dataset, PASCAL VOC 2007 dataset, ImageNet dataset, VQA dataset
- ▶ *Grad-CAM++: Improved Visual Explanations for Deep Convolutional Networks*: ImageNet (ILSVRC2012) dataset, PASCAL VOC 2007 dataset, PASCAL VOC 2012 dataset, CIFAR-10 dataset, Flickr30k dataset, Sports-1M dataset
- ▶ *Score-CAM: Gradient-free visual explanations*: ImageNet (ILSVRC2012) dataset, PASCAL VOC dataset

Project Datasets

- ▶ *Testing on small-resolution images:* **CIFAR-10 dataset**
- ▶ *Testing on large-resolution images:* **Fruits-360 dataset**

Research Gaps Identified

- ▶ **Absence of Grad-CAM (or any XAI) for CNNs Trained on Frequency Domain Inputs:** None explained how to interpret or explain the predictions made by these frequency-domain-trained CNNs using XAI methods like Grad-CAM.
- ▶ **Lack of Mapping Explanations from Frequency Domain Features to Original Spatial Images:** No existing research has looked into using Grad-CAM on frequency-domain feature maps from CNNs that were trained using frequency-domain data. The main issue is figuring out how to convert these frequency-based explanations back into the original spatial images to find which pixels contributed to the model's decisions.

Research Gaps Identified

- ▶ **Unexplored Interpretability Challenges for Frequency-Domain Architectures:** The interpretability of models that operate primarily or entirely in the frequency domain is not addressed. How do specific frequency components, identified as "important" by an XAI method, translate to human-understandable visual features in the original image?

Problem Statement

How can the predictions made by a CNN that was trained and tested using images in the frequency domain be clearly explained in terms of the specific areas in the original spatial images, using Grad-CAM/Score-CAM and explainable AI techniques?

Objectives

1. Adapting Score-CAM for Convolutional Neural Networks (CNNs) Operating in the Frequency Domain.
2. Creating a strong approach to connect frequency-domain Grad-CAM explanations back to the original image in the spatial domain.
3. Evaluating how clear and accurate the generated spatial explanations are?, both in terms of the amount of information provided and the quality of that information.
4. Studying how specific aspects in the frequency domain affect the ability to understand spatial information?

Proposed Methodology

1. Dataset Preparation and Preprocessing
2. Frequency Domain Transformation using FFT
3. Frequency Domain CNN Architecture (FDCNN)
4. Training Strategy and Optimization
5. Score-CAM Explainability
6. Mapping to Spatial Domain

1. Dataset Preparation and Preprocessing

Dataset: Fruits-360 (100×100 RGB images)

Data Augmentation (Training):

- ▶ Random flips (horizontal: $p = 0.5$, vertical: $p = 0.3$)
- ▶ Random rotation: $[-20, +20]$
- ▶ Color jittering: brightness, contrast, saturation ($\pm 20\%$)
- ▶ Normalization: $\mu = [0.485, 0.456, 0.406]$,
 $\sigma = [0.229, 0.224, 0.225]$

Data Split:

- ▶ Training: 85% | Validation: 15% | Test: Separate set

2. Frequency Domain Transformation using FFT

2D Fast Fourier Transform:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I(x, y) \cdot e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})} \quad (1)$$

Magnitude and Phase Extraction:

$$M(u, v) = |F(u, v)|, \quad \Phi(u, v) = \arg(F(u, v)) \quad (2)$$

Log-scale Normalization:

$$M_{norm}(u, v) = \frac{\log(M(u, v) + \epsilon) - \mu_{M_{log}}}{\sigma_{M_{log}} + 10^{-8}} \quad (3)$$

Phase Encoding:

$$\Phi_{cos}(u, v) = \cos(\Phi(u, v)), \quad \Phi_{sin}(u, v) = \sin(\Phi(u, v)) \quad (4)$$

9-Channel Feature Concatenation:

$$F_{features} = [M_{norm}^R, M_{norm}^G, M_{norm}^B, \Phi_{cos}^R, \Phi_{cos}^G, \Phi_{cos}^B, \Phi_{sin}^R, \Phi_{sin}^G, \Phi_{sin}^B] \quad (5)$$

3. Frequency Domain CNN Architecture (FDCNN)

Base Architecture: Modified ResNet50

Input Layer Modification:

- ▶ Original: Conv2d(3, 64, kernel_size=7)
- ▶ Modified: Conv2d(9, 64, kernel_size=7) (*9 frequency channels*)

Residual Block:

$$y = \mathcal{F}(x, \{W_i\}) + x \quad (6)$$

Enhanced Classification Head:

Global Average Pooling (2048)

↓ Dropout(0.5) ↓ Linear(2048→1024) + ReLU + BN

↓ Dropout(0.35) ↓ Linear(1024→512) + ReLU + BN

↓ Dropout(0.25) ↓ Linear(512→*num_classes*)

4. Training Strategy and Optimization

Loss Function (Cross-Entropy with Label Smoothing):

$$\mathcal{L}_{CE} = - \sum_{i=1}^C y'_i \log(\hat{y}_i), \quad y'_i = \begin{cases} 1 - \alpha & \text{if } i = \text{true class} \\ \frac{\alpha}{C-1} & \text{otherwise} \end{cases} \quad (7)$$

where $\alpha = 0.1$

AdamW Optimizer with Differential Learning Rates:

- ▶ Pretrained layers: $\eta = 0.0001$, New layers: $\eta = 0.001$
- ▶ Weight decay: $\lambda = 5 \times 10^{-4}$

Cosine Annealing with Warm Restarts:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min}) \left(1 + \cos \left(\frac{T_{cur}}{T_i} \pi \right) \right) \quad (8)$$

Regularization:

- ▶ Gradient clipping (max norm = 1.0)
- ▶ Mixed precision training (FP16/FP32)
- ▶ Early stopping (patience = 15 epochs)

5. Score-CAM Explainability

Step 1: Extract and Normalize Activation Maps

$$A_{norm}^k = \frac{A_{up}^k - \min(A_{up}^k)}{\max(A_{up}^k) - \min(A_{up}^k)}, \quad k = 1, \dots, 2048 \quad (9)$$

Step 2: Generate Masked Inputs and Compute Scores

$$I_{masked}^k = I \odot A_{norm}^k, \quad S_c^k = \text{Softmax}(f(I_{masked}^k))_c \quad (10)$$

Step 3: Compute Importance Weights

$$w^k = \frac{S_c^k}{\max_j S_c^j} \quad (11)$$

Step 4: Generate Score-CAM

$$\text{CAM}_c = \text{ReLU} \left(\sum_{k=1}^{2048} w^k \cdot A^k \right) \quad (12)$$

6. Mapping Score-CAM to Spatial Domain

Step 1: Mask Frequency Magnitude

$$M'(u, v) = M_{norm}(u, v) \odot CAM_{freq}(u, v) \quad (13)$$

Step 2: Reconstruct Complex Representation

$$F'(u, v) = M'(u, v) \cdot e^{j\Phi(u, v)} \quad (14)$$

Step 3: Apply Inverse FFT

$$I_{cam}(x, y) = \text{IFFT2D}(\text{IFFTShift}(F'(u, v))) \quad (15)$$

Saliency Map Refinement Pipeline:

- ▶ Extract absolute value and average RGB channels
- ▶ Invert for fruit focus: $S_{inv} = \max(S_{gray}) - S_{gray}$
- ▶ Gaussian smoothing ($\sigma = 2.5$) + Normalization
- ▶ Adaptive thresholding (40th percentile)
- ▶ Morphological operations (closing + opening)
- ▶ Contrast enhancement: $S_{final} = \left(\frac{S_{masked}}{\max} \right)^{0.7}$

7. Visualization and Overlay Generation

Adaptive Alpha Blending:

$$\alpha(x, y) = 0.7 \cdot S_{final}(x, y) \quad (16)$$

$$I_{highlight}(x, y) = (1 - \alpha) \cdot I_{orig} + \alpha \cdot C_{heat} \quad (17)$$

Eight-Panel Comprehensive Display:

1. Original Image with ground truth
2. Frequency Domain (magnitude spectrum)
3. Score-CAM in frequency domain
4. Prediction with confidence score
5. Spatial saliency map
6. Highlighted important regions
7. Importance heatmap overlay (50%)
8. Before/After comparison

Why my approach is better?

- ▶ **Why did I use Score-CAM instead of Grad-CAM?**

Score-CAM is better than Grad-CAM because it does not rely on gradients, which can be noisy or vanish, and instead uses the actual increase in class score caused by masking the image with each activation map. This makes the explanations gradient-free, more stable, less noisy, and more reliable.

- ▶ **Why is only the magnitude of the frequency domain image not sufficient for better explainability and visualization?**

Magnitude alone isn't enough because phase carries the structural details of the image. Without phase, the spatial layout is lost, leading to poor explainability and misleading visualizations.

Why my approach is better?

- ▶ **Why do only the magnitude and one phase also lead to poor visualizations and explainability?**

Using magnitude with a single fixed phase fails because phase encodes the unique spatial structure of each image. A constant phase makes different images look similar in the frequency domain, leading to wrong features, wrong predictions, and poor explainability.

- ▶ **How did choosing magnitude and two phases help to improve the explainability?**

Using magnitude with both phase components works because it restores the full phase information, which preserves each image's unique spatial structure. This gives distinct frequency representations for different images, leading to correct features, predictions, and visualizations.

Implementation Details

Core Technologies

- ▶ **Framework:** PyTorch 1.12+
- ▶ **Language:** Python 3.8+
- ▶ **Architecture:** ResNet50 (Modified)
- ▶ **Computer Vision:** TorchVision
- ▶ **Numerical Computing:** NumPy, SciPy
- ▶ **Visualization:** Matplotlib

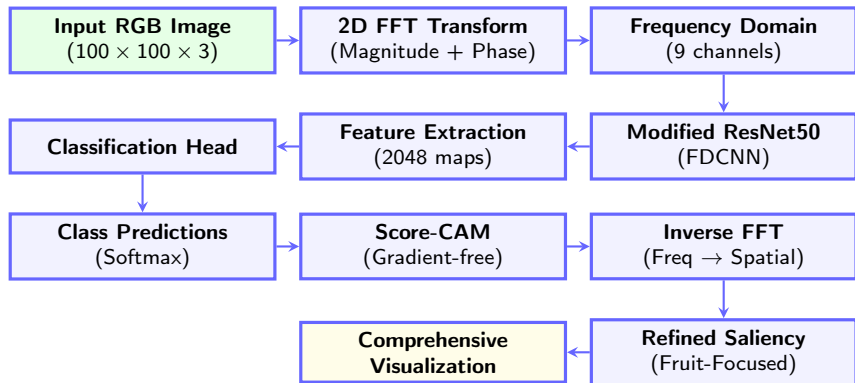
Hardware Specifications

- ▶ **GPU:** 16GB VRAM
- ▶ **Batch Size:** 64
- ▶ **Precision:** Mixed (FP16/FP32)
- ▶ **Workers:** 4 parallel loaders
- ▶ **Optimization:** CuDNN benchmark
- ▶ **Memory:** Pin memory enabled

Key Feature

GPU-accelerated FFT operations via `torch.fft` for efficient frequency domain transformations

Complete Pipeline Summary



Pipeline Flow: Input → FFT → Frequency CNN → Classification → Score-CAM → Spatial Mapping → Visualization

Fruits-360 Visualization Results

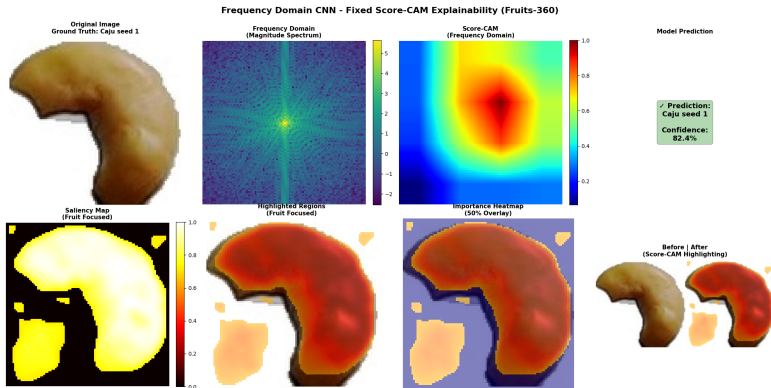


Figure 1: Fruits-360 Score-CAM explainability visualization showing frequency domain analysis and fruit-focused spatial saliency mapping for Caju seed classification.

Fruits-360 Training Results

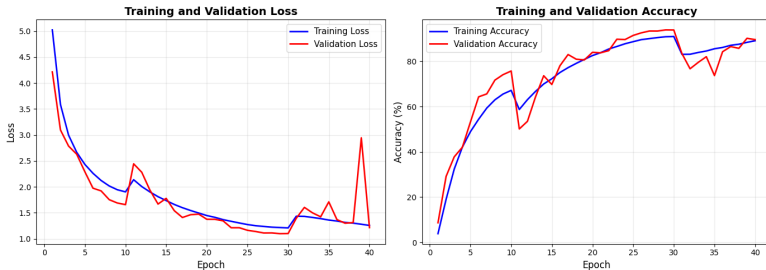


Figure 2: Fruits-360 training and validation curves over 40 epochs showing loss convergence and accuracy progression with cosine annealing warm restart.

CIFAR-10 Visualization Results

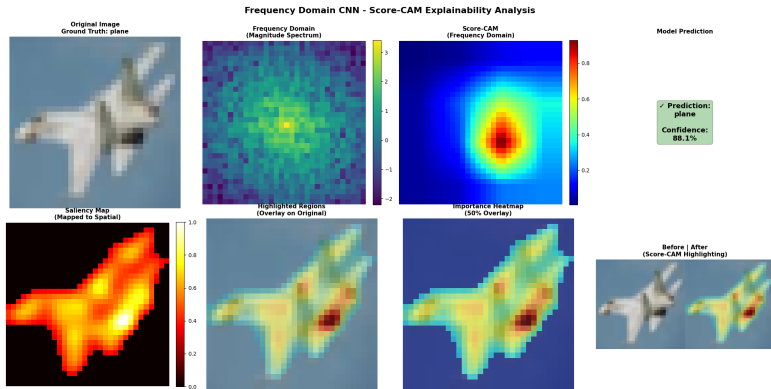


Figure 3: CIFAR-10 Score-CAM explainability visualization showing frequency domain transformation, activation mapping, and spatial saliency localization for airplane classification.

CIFAR-10 Training Results

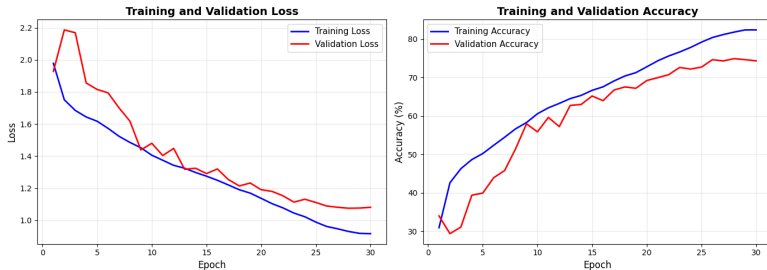


Figure 4: CIFAR-10 training and validation curves showing loss convergence and accuracy improvement over 30 epochs.

Conclusion

- ▶ This project effectively tackles the main problem of explaining how CNNs make predictions when they're trained using frequency domain data by translating those predictions back into visual images that are easier to understand. The research creates a full approach that links image classification in the frequency domain with clear and explainable AI using the Score-CAM method.

Key Achievements

- ▶ Successfully developed **FDCNN with Score-CAM** framework connecting frequency domain classification with explainable AI through spatial mapping
- ▶ Achieved **82.78% accuracy** on Fruits-360 dataset (225 classes, 39,212 test samples) with Cohen's Kappa of 0.8270
- ▶ Implemented gradient-free Score-CAM providing stable, interpretable class activation maps without backpropagation
- ▶ Developed novel inverse FFT pipeline with multi-step refinement producing fruit-focused spatial saliency maps

Research Contributions

- ▶ **Theoretical:** Demonstrated frequency domain representations with log-scale magnitude and trigonometric phase encoding provide discriminative features for visual classification
- ▶ **Methodological:** Established complete end-to-end framework from FFT transformation through classification to interpretable spatial explainability visualization

Limitations

- ▶ Computational Overhead
- ▶ Score-CAM Efficiency
- ▶ Spatial Resolution
- ▶ Domain Specificity

Future Work Objectives and Motivation

Hybrid Spatial-Frequency Architecture

- ▶ **Objective:** Create a single system that handles both spatial and frequency information at the same time, using the best features of each to improve performance.
- ▶ **Motivation:** Right now, most work processes focus only on the frequency domain. Spatial features help in understanding local patterns and object meanings, whereas frequency features are better at capturing the overall structure and texture of an image. By combining these two approaches, we can achieve better results.

Future Work Expected Outcomes

- ▶ 5-10% accuracy improvement over single-domain approaches
- ▶ More robust predictions by leveraging complementary features
- ▶ Enhanced explainability showing both spatial and frequency contributions
- ▶ Better handling of different image characteristics (texture-rich vs. shape-dominated)

Future Work Challenges

- ▶ Managing increased computational cost of dual-stream processing
- ▶ Designing effective attention mechanisms for cross-domain feature interaction
- ▶ Balancing learning between spatial and frequency streams to prevent dominance
- ▶ Extending Score-CAM to visualize contributions from both domains simultaneously

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